Experimental Study on Spurious Suppression Techniques of S-band Magnetrons

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***Abstract*—Magnetrons are good candidates for large power microwave sources in various industrial applications. They are unstable oscillators with frequency and amplitude fluctuations, which easily cause electromagnetic interference (EMI) problems. Two regular methods to improve magnetrons' spectrum purity are compared in experiments using 1-kW S-band cook-type magnetrons. One method is to depress anode voltage ripple, which reduces the output frequency fluctuation range to 7.9%. The other method is injection-locking, which reduces the range to 1%. The injection-locking technique is more suitable for applications with requirements of high-frequency stability and controllability. Only low spurious noise and high-frequency purity are required in a few industrial applications. It is an economically feasible approach to depressing the voltage ripple by reducing filament current and adding a capacitor-based filter module. The study provides a reference to suppressing the EMI caused by magnetrons in different demands and scenarios.**

Keywords—magnetrons, voltage ripple, output spectrum, electromagnetic interference, electromagnetic compatibility

I. INTRODUCTION

Nowadays, microwave has been widely used in various fields of military and civilian use, such as microwave heating, microwave communication, and microwave wireless power transmission.[1][2][3] Magnetrons attracted great interest as a microwave source because of the advantages of high efficiency, compact size, and low cost[4]. But free-running magnetrons always have random phase fluctuations and wide spectral bands. The poor output quality of magnetrons not only restricts their applications in fields that require precious control but makes electrical devices operating in the same frequency easily affected; for example, Zigbee sensors, Bluetooth devices, or Wi-Fi devices are at high risk of electromagnetic interfered from S-band magnetrons.

A practical method to improve the output spectrum purity of a magnetron is injection locking. B. Yang et al. built a simultaneous wireless information and power transfer system based on a phased array using four 5.8-GHz injection-locked magnetrons [5]. Chen et al. proposed a magnetron with a solid-source-like performance by injection-locking and close-loop compensation [6]. Another improvement idea is to make power supplies better. Mitani et al. proposed turning off the filament current to improve the output quality of a magnetron [7]. Tsong-Shing Lee designed a circuit combined filament-heating and cavity-driven circuit with gain-frequency regulation control and achieved better constant power operation [8]. Chen et al. improved the output of a 20kW magnetron by reducing the anode voltage ripple of a magnetron and obtained better injection-locking performance [9]. However, the injection-locking technique needs a waveguide isolator to provide an injection loop, and the heavy and expensive isolators make injection-locking incompatible for some applications.

In this paper, an experimental study on injection-locking and voltage ripple suppression is carried out. A switch-mode DC power supply is improved to enhance the output purity of an S-band magnetron, and an injection-locking experiment on the magnetron is conducted. The investigation also compares the two improvement methods, which will provide a reference for choosing a suitable way to reduce the spurious noise of an S-band magnetron in different applications.

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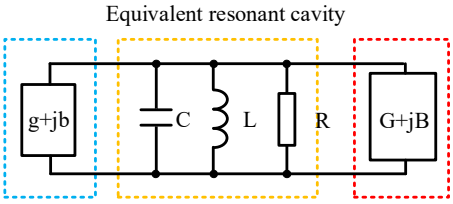


Fig. 1. Equivalent circuits of a magnetron's resonant cavity.

II. ANALYSIS OF MAGNETRONS

A. Output characteristic of magnetrons

A magnetron is always considered as a one-port oscillator, and the equivalent circuit can be described as the form shown in Fig.1. The equivalent circuit includes the equivalent electronic admittance $g+jb$, the equivalent oscillating circuit with parameters L, C, and R, the equivalent load admittance $G+jB$. When the magnetron satisfies the single-mode oscillation condition, the circuit equation can be written as:

$$g + jb = \frac{1}{R} + j\omega_c C + \frac{1}{j\omega_c L} + G + jB \qquad (1)$$

The operating frequency ω_c is one of the most critical parameters of microwave sources. According to J. C. Slater's work [10], the equivalent electronic admittance of the magnetron is given as follows:

$$g = \frac{1}{R}\left(\frac{V_{dc}}{V_{RF}} - 1\right) \qquad (2)$$

$$b = b_0 + g\tan\alpha \qquad (3)$$

where V_{dc} is the voltage between the anode and cathode of magnetrons, V_{RF} is the amplitude of the voltage of the

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magnetron's actual output, b_0 is a constant, α is defined as the pushing parameter, its unit is rad, and its value varies with the structure of magnetrons. Combining (1) – (3), the output frequency ω of a magnetron is obtained:

$$\omega = \omega_0 + \frac{b_0 + B}{2C} - \frac{\tan\alpha}{2Q_0}\left(\frac{V_{DC}}{V_{RF}} - 1\right) \qquad (4)$$

The output characteristic of a magnetron is mainly affected by its anode voltage V_{dc}. Thus, reducing the ripple of the anode voltage is important for improving the output quality of a magnetron.

B. *Injection-locking technique*

The injection-locking technique is to inject a low-power and high-stability external signal into a high-power continuous wave magnetron. When the frequency and power of the external signal satisfy certain conditions, the output frequency and phase of the magnetron's output are synchronous with the injected signal. Besides, the magnetron's output quality will be improved significantly, frequency fluctuations of the magnetron can be suppressed, and a purer output spectrum than free-running is obtained. In 1947, Adler et al. studied the injection-locking theory of microwave oscillators based on small signal gain approximation and theoretically deduced the injection-locked condition of an oscillator [11]:

$$\frac{2Q_{ext}(\omega_1 - \omega)}{\omega} \le \rho \qquad (5)$$

where ω_1 is the angular frequency of the external signal, $\rho = \sqrt{P_1 / P_0}$ is the injection ratio, P_1 is the power of the external signal, P_0 is the output power of the magnetron, and Q_{ext} is the external quality factor of the magnetron. (5) gives out the condition of an injection-locked magnetron needs to be satisfied and provides the theoretical basis of experiment study on the injection-locking of a magnetron.

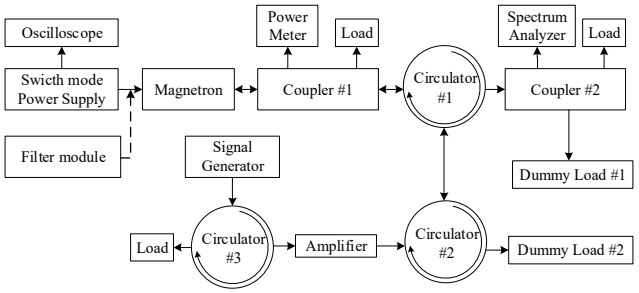
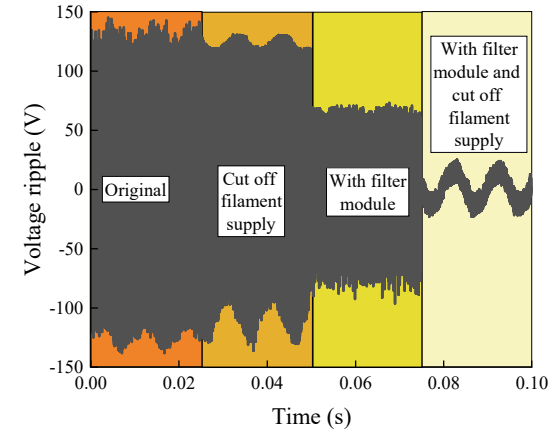


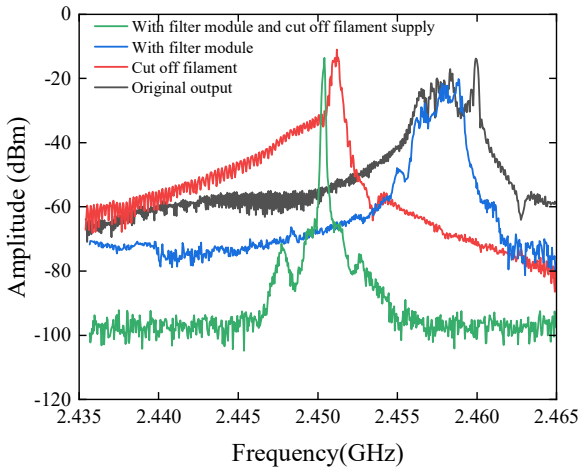
Fig. 2. Diagram of the experiment system.

III. Experiment and Results

The experiment system diagram is shown in Fig. 2. The magnetron is Panasonic, 2M210-M1, working at 2.45GHz. A signal source (Hitter HMC-T2220) and a high gain power amplifier (mini-circuits ZHL-30W-262-S+) generate the injected external signal. The magnetron output power is measured by an AV2433 power meter, and the spectrum is detected by a spectrum analyzer (FSP, R&S). The anode voltage waveform of the magnetron is measured by the Tektronix DPO7254 oscilloscope. In this experiment, the output power is 900W. The power supply is Magmeet WELAMP 2000F.



(a)



(b)

Fig. 3. Results of the power supply improvement experiments. (a) Voltage ripples in different conditions. (b) Output spectra of the magnetron in other states.

A. *Improvement of voltage ripple*

The switch mode power supply will provide a DC voltage up to 4200V, driving the magnetron. Due to the imperfect filter, the switching frequency of the MOS tube used in the power supply will exist and deteriorate the output purity of the magnetron. Besides, the filament heating supply will increase the voltage ripple and make the output characteristic more scattered. This experiment uses a 0.14μF capacitor filter module to suppress the voltage ripple. And when the magnetron operation is stable, the filament supply will be cut off. We compare the voltage ripples of the power supply and the output spectra of the magnetron in four conditions: 1) Original voltage ripple; 2) Cut off the filament heating supply; 3) With the capacitor filter module; 4) With the capacitor filter module and cut off the filament heating supply. Fig. 3 gives out the results.

The voltage ripple of the switch mode power supply is shown in Fig. 3 (a). We know the peak-peak value of the original voltage ripple is 283V. The spectrum width is defined as the frequency bandwidth when the amplitude of the output spectrum is one-hundredth of the maximum value of 4.43MHz. When the filament supply is cut off, the peak-peak value of the voltage ripple is almost the same, but as shown in Fig. 3 (a), the purity of the voltage ripple is improved. The spectrum width is 1.48MHz, about one-third

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of the original output. The voltage ripple of the power supply with the capacitor filter module is 174V, about half of the original value, and the spectrum width is 3.39MHz. This result indicates that even the capacitor filter module has a better effect on reducing voltage ripple than cutting off the filament supply, but cutting off the filament supply is more effective in improving the output purity of a magnetron. Moreover, the output state of the magnetron is not stable. Only adding a capacitor filter module at the output port of the DC power supply may deteriorate the magnetron's output. The best spurious suppression result is cutting off the filament current while adding the capacitor filter module. The voltage ripple of the power supply is suppressed to 48V (peak-peak value), and the spectrum width is 0.35MHz. The purity of the magnetron's output spectrum has significantly improvement.

B. Injection-locking experiment

The injection-locking technique is a reliable method to enhance the output purity of magnetrons. An injection-locking experiment is carried out with a power of the injection signal of 9W and the injection ratio ρ=0.1. This experiment is conducted under the same four conditions as III. A. The result of the investigation is shown in Fig. 4. We know when the magnetron is injection-locked, its output characteristic will be improved remarkably. The spectrum widths of the original output, cut-off filament supply, with filter module and with filter module, and cut-off filament supply are 0.65MHz, 0.12MHz, 0.08MHz, and 0.05MHz, respectively. Besides, the spurious noise in the measurement frequency range is reduced markedly. Compared with Fig. 3 (a), we found that even with the injection-locking technique, the output purity without any ripple improvement method is not as good as a free-running state when adding a filtering capacitor and cutting off the filament supply. It is worth mentioning that injection-locking improves the unstable operating state only with the capacitor filter module. The magnetron has a purer and better controllability than only cutting off the filament supply, which has a superior output performance in a free-running state.

The injection-locking technique makes the magnetrons' output controllable and reduces the output noise. However, an injection-locked magnetron system is expensive and large. According to the experiment results, cutting off the filament supply and adding a capacitor filter module may be a more economical method in some industrial applications without precise control. The improvement of output purity is better than injection-locking the magnetron without any voltage ripple enhancement.

IV. CONCLUSION

In this work, the study on improving the output spectrum purity of S-band magnetrons is carried out, and two conventional methods are compared for different application scenarios.

The output spectrum is improved in either method. The spectrum width of the magnetron is reduced to 0.35MHz by voltage ripple suppression, which is better than using the injection-locking technique directly. The injection-locking technique is an effective and irreplaceable method for applications sensitive to controllable, stable, and low-interference sources, such as power sources in microwave communication and particle accelerators. For those who only need a relatively stable output, such as industrial microwave heating, suppressing the voltage ripple of magnetrons is an economical and low-cost approach. This experimental study provides a reference for choosing a better way to reduce the magnetrons' electromagnetic interference.

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